Elementary Analysis

Section 21: More on Metric Spaces: Continuity

Exercises 21.1–21.14, with every subpart
Detailed worked solutions

Mathjock

Text and numbering. Kenneth A. Ross, *Elementary Analysis: The Theory of Calculus*, second edition (Springer, 2013), Sections 21–38. Exercise numbers follow the copy already linked on Mathjock. These are independently written worked solutions, not the publisher's solutions manual. Each exercise is identified by a short mathematical description.

Reading philosophy. Every argument states its goal, records the assumptions being used, and explains the intermediate algebra or logical inference. Definitions, sign conditions, and both directions of an "if and only if" are made explicit. A cited rule is named at the point where it is used.

Conventions. $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, and $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$. Unless a different domain is stated, sequence indices run through $\mathbb{N}$ and all terms and finite limits are real. The empty sum is 0, and $0! = 1$.

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21 More on metric spaces: continuity

A metric d measures distance. The open ball $B_r(s)$ consists of the points whose distance from s is less than r. A continuous function sends points sufficiently close to a *fixed* point to points close to its image. Uniform continuity uses one input tolerance for all pairs of points. An open set in a subspace is understood relative to that subspace; it need not be open in the surrounding Euclidean space.

21.1 Exercise 21.1: Uniform continuity of a vector-valued function

What we must show. Show that $\gamma(t) = (f_1(t), \ldots, f_k(t))$ is uniformly continuous when all its coordinate functions are uniformly continuous.

Plan. Make each coordinate error small enough that the sum of their squares is less than the desired squared distance.

Choose the input tolerance before choosing the points.

Fix $\varepsilon > 0$. For each j, uniform continuity of f_j gives $\delta_j > 0$ such that, for *all* $s, t \in [a, b]$,

$$|s - t| < \delta_j \implies |f_j(s) - f_j(t)| < \varepsilon/\sqrt{k}.$$

The minimum $\delta = \min(\delta_1, \ldots, \delta_k)$ is positive because there are only finitely many coordinates. If $|s - t| < \delta$, every coordinate estimate holds. Consequently

$$d(\gamma(s), \gamma(t)) = \left(\sum_{j=1}^{k} |f_j(s) - f_j(t)|^2 \right)^{1/2} < \left(k \frac{\varepsilon^2}{k} \right)^{1/2} = \varepsilon.$$

Conclusion. The same δ works for every pair s, t, which is precisely uniform continuity.

21.2 Exercise 21.2: Continuity at one point in terms of open sets

What we must show. Prove both directions of the open-neighborhood characterization at $s_0 \in S$.

Continuity implies the neighborhood condition.

Suppose f is continuous at s_0 and $U \subseteq S^*$ is open with $f(s_0) \in U$. Openness provides $\varepsilon > 0$ for which $B^*_\varepsilon(f(s_0)) \subseteq U$. Continuity supplies $\delta > 0$ with $d(s, s_0) < \delta$ implying $d^*(f(s), f(s_0)) < \varepsilon$. The open set $V = B_\delta(s_0)$ contains s_0 and satisfies $f(V) \subseteq U$.

The neighborhood condition implies continuity.

Conversely, fix $\varepsilon > 0$ and apply the assumed condition to the open ball $U = B^*_\varepsilon(f(s_0))$. We obtain an open V containing s_0 with $f(V) \subseteq U$. Since V is open, it contains a ball $B_\delta(s_0)$ for some $\delta > 0$. Hence

$$d(s, s_0) < \delta \implies s \in V \implies f(s) \in U \implies d^*(f(s), f(s_0)) < \varepsilon.$$

Conclusion. This is exactly the metric definition of continuity at s_0.

21.3 Exercise 21.3: Distance from a fixed point is uniformly continuous

What we must show. For fixed s_0, prove uniform continuity of $f(s) = d(s, s_0)$.

Derive the reverse triangle inequality.

The triangle inequality gives $d(s, s_0) \leq d(s, t) + d(t, s_0)$, so $d(s, s_0) - d(t, s_0) \leq d(s, t)$. Interchanging s, t bounds the negative of the same difference. Together these statements give

$$|f(s) - f(t)| = |d(s, s_0) - d(t, s_0)| \leq d(s, t).$$

Given $\varepsilon > 0$, choose $\delta = \varepsilon$. Whenever $d(s, t) < \delta$, the displayed inequality implies $|f(s) - f(t)| < \varepsilon$.

Conclusion. Distance from a fixed point is even Lipschitz with constant 1; this means its output distance never exceeds its input distance.

21.4 Exercise 21.4: Rational intervals suffice to test continuity

What we must show. Establish the equivalence of continuity, openness of preimages of all open intervals, and openness for intervals with rational endpoints.

The first two implications.

If f is continuous, Theorem 21.3 makes the inverse image of every open subset of $\mathbb{R}$ open in S. In particular, this holds for every interval (a, b). Restricting a, b to rationals gives the third condition.

Recover continuity using only rational endpoints.

Assume $f^{-1}((a, b))$ is open whenever $a < b$ are rational. Fix $s_0 \in S$ and $\varepsilon > 0$. Density of $\mathbb{Q}$ gives rationals with

$$f(s_0) - \varepsilon < a < f(s_0) < b < f(s_0) + \varepsilon.$$

The set $V = f^{-1}((a, b))$ is open and contains s_0, so some $B_\delta(s_0)$ is contained in V. If $d(s, s_0) < \delta$, then $a < f(s) < b$, and therefore $|f(s) - f(s_0)| < \varepsilon$.

Conclusion. Continuity at every s_0 follows. A dense set of possible interval endpoints gives arbitrarily small neighborhoods of every output.

21.5 Exercise 21.5: What can fail on a noncompact Euclidean set

What we must show. Construct (a) an unbounded continuous function and (b) a bounded continuous function with no maximum on a noncompact $E \subseteq \mathbb{R}^k$.

Plan. By Heine–Borel, a compact Euclidean set is exactly a closed bounded set. Failure of compactness gives either points escaping to infinity or a missing limit point.

(a) Construct an unbounded function.

If E is unbounded, set $h(x) = \|x\|$, the Euclidean distance from zero. Exercise 21.3 proves continuity. For each $M > 0$, unboundedness of E supplies $x \in E$ with $h(x) > M$.

If E is bounded, it cannot be closed. Choose $x_0 \in \overline{E} \setminus E$ and set $h(x) = 1/\|x - x_0\|$ on E. The denominator is positive on E, and the distance function and the reciprocal function on $(0, \infty)$ are

continuous, so h is continuous. Since $x_0 \in \overline{E}$, for every $M > 0$ some $x \in E$ satisfies $0 < \|x - x_0\| < 1/M$; thus $h(x) > M$.

(b) Compress the unbounded values into a bounded interval.

In either case $h \geq 0$ and h is unbounded above. Define

$$g(x) = \frac{h(x)}{1 + h(x)} = 1 - \frac{1}{1 + h(x)}.$$

This is continuous and $0 \leq g(x) < 1$. For any $x \in E$, choose $y \in E$ with $h(y) > h(x)$. Since $t/(1+t)$ is strictly increasing for $t \geq 0$ (as follows by cross multiplication), $g(y) > g(x)$. Thus no point can maximize g.

Conclusion. Compactness is essential both for boundedness of continuous functions and for attainment of their extreme values.

21.6 Exercise 21.6: Compositions of continuous metric-space functions

What we must show. Show that $g \circ f : S_1 \to S_3$ is continuous when both component functions are continuous.

Pull an open set back in two stages.

Let $U \subseteq S_3$ be open. Continuity of g says $g^{-1}(U)$ is open in S_2. Continuity of f then says $f^{-1}(g^{-1}(U))$ is open in S_1. The identity

$$(g \circ f)^{-1}(U) = f^{-1}(g^{-1}(U))$$

holds because each side consists of exactly those s with $g(f(s)) \in U$. Theorem 21.3 therefore proves continuity of $g \circ f$.

Conclusion. The order of inverse images is the reverse of the order in which the functions are applied.

21.7 Exercise 21.7: Restricted metrics and continuity of a path

(a) Restricting a metric.

For $s, t, u \in E \subseteq S$, the numbers $d(s,t)$ are nonnegative, $d(s,t) = 0$ exactly when $s = t$, symmetry holds, and $d(s,u) \leq d(s,t) + d(t,u)$. These statements already hold in S, so they still hold when the available points are restricted to E. Thus (E, d) is a metric space. In particular $d(x,y) = |x - y|$ is a metric on every subset of $\mathbb{R}$.

(b) Spell out the definition for a path.

A function $\gamma : [a,b] \to \mathbb{R}^k$ is continuous if, for every $t_0 \in [a,b]$ and every $\varepsilon > 0$, there is $\delta > 0$ such that

$$t \in [a,b], \quad |t - t_0| < \delta \quad \Longrightarrow \quad \left(\sum_{j=1}^{k} |\gamma_j(t) - \gamma_j(t_0)|^2 \right)^{1/2} < \varepsilon.$$

Conclusion. At an endpoint, only points of $[a,b]$ are tested. The definition does not ask for values of the path outside its domain.

21.8 Exercise 21.8: Uniform continuity sends Cauchy sequences to Cauchy sequences

What we must show. Prove that $(f(s_n))$ is Cauchy in S^* when (s_n) is Cauchy in S and f is uniformly continuous.

Match the two definitions.

Fix $\varepsilon > 0$. Uniform continuity gives $\delta > 0$ such that $d(s,t) < \delta$ implies $d^*(f(s), f(t)) < \varepsilon$, for all $s, t \in S$. Since (s_n) is Cauchy, some N has $d(s_m, s_n) < \delta$ whenever $m, n > N$. Consequently

$$m, n > N \implies d^*(f(s_m), f(s_n)) < \varepsilon.$$

Conclusion. This is the Cauchy condition in the target. We did not assume either sequence has a limit. Mere continuity would not suffice: $1/x$ on $(0,1)$ sends the Cauchy sequence $1/(n+1)$ to the non-Cauchy sequence $n+1$.

21.9 Exercise 21.9: Continuous surjections involving a square

(a) Project the square onto an interval.

On $[0,1]^2$, put $p(x_1, x_2) = x_1$. The inequality $|p(x) - p(y)| \le \|x - y\|$ proves continuity. Every $u \in [0,1]$ is the image of $(u, 0)$, so the range is exactly $[0,1]$.

(b) Can an interval fill a square?

Yes: a continuous surjection $[0,1] \to [0,1]^2$ is called a *space-filling curve*. Dimension alone does not prevent one. The question asks for a judgment, rather than a formula; here is an existence argument explaining why the answer is not merely a guess.

Let C be the middle-thirds Cantor set. Each point of C has a unique expansion $t = \sum_{j \ge 1} 2b_j 3^{-j}$ with $b_j \in \{0, 1\}$. Define

$$H(t) = \left(\sum_{j \ge 1} b_{2j-1} 2^{-j},\ \sum_{j \ge 1} b_{2j} 2^{-j} \right).$$

Any two numbers in $[0,1]$ have binary expansions; interlacing their digits produces a point of C with precisely those two images. Thus $H(C) = [0,1]^2$. If two Cantor points have their first $2m$ ternary digits equal, their coordinate differences are at most $\sum_{j>m} 2^{-j} = 2^{-m}$. Conversely, points in distinct cylinders specified by the first $2m$ digits are separated by a positive gap. These finite-cylinder gaps prove continuity of H on C.

Extend H to each removed interval (u, v) by straight-line interpolation between $H(u)$ and $H(v)$. The resulting map stays in the square because line segments between square points stay in the square. It is continuous inside each gap. To check continuity at $t \in C$, consider any sequence $t_n \to t$. For points in C use continuity of H. For points in gaps, either a subsequence lies in a fixed gap whose endpoint is t (where linear interpolation is continuous), or the gap lengths tend to zero. In the latter case both endpoints tend to t, and their images tend to $H(t)$, so every convex combination does too. Only finitely many gaps have length at least any prescribed positive number; hence these cases cover every possible subsequence. The sequential criterion proves continuity.

Conclusion. The extension is a continuous map on $[0,1]$ and is already onto the square on its Cantor subset. A curve can therefore fill a square.

21.10 Exercise 21.10: Four questions about continuous ranges

(a) Map $(0,1)$ onto $[0,1]$.

Use $f(t) = (1 + \sin(4\pi t))/2$. This continuous function takes values only in $[0,1]$, attains 1 at $t = 1/8$, and attains 0 at $t = 3/8$. The intermediate value theorem on $[1/8, 3/8]$ supplies every value between them. Hence the map is onto.

(b) Map $(0,1)$ onto $\mathbb{R}$.

Use $g(t) = \tan(\pi(t - 1/2))$. Its argument lies in $(-\pi/2, \pi/2)$, where tangent is continuous and strictly increasing, with limits $-\infty$ and $+\infty$ at the two ends. For any real y, choose interior points where g is below and above y and apply the intermediate value theorem.

(c) Map $[0,1] \cup [2,3]$ onto $[0,1]$.

Define $h(t) = t$ on $[0,1]$ and $h(t) = t - 2$ on $[2,3]$. On each component this is continuous. A ball of radius less than 1 around any domain point cannot reach the other component, so continuity on each component proves continuity on the whole domain. Each component already maps onto $[0,1]$.

(d) Rule out two proposed ranges.

A continuous image of the compact interval $[0,1]$ is compact. Neither $(0,1)$ (not closed in $\mathbb{R}$) nor $\mathbb{R}$ (unbounded) is compact. If a continuous map were onto either, its image would be both compact and noncompact, a contradiction.

Conclusion. The formulas supply (a)–(c); preservation of compactness excludes both maps in (d).

21.11 Exercise 21.11: The rationals are not a countable intersection of open sets

What we must show. Use the Baire theorem to rule out $\mathbb{Q} = \bigcap_{n \geq 1} U_n$ with all U_n open in $\mathbb{R}$.

Plan. Such U_n would all be dense. Intersect them also with the complements of individual rationals and reach an empty dense set.

Construct dense open sets.

Suppose the representation exists. Since every U_n contains $\mathbb{Q}$ and $\mathbb{Q}$ is dense, U_n is dense. List the rationals as $\mathbb{Q} = \{q_1, q_2, \ldots\}$. Each $V_n = \mathbb{R} \setminus \{q_n\}$ is open and dense: every nonempty open interval contains points other than a single fixed point.

Apply completeness exactly where needed.

The real line is a complete metric space. Theorem 21.7(a), applied to the interlaced sequence $U_1, V_1, U_2, V_2, \ldots$, says their intersection is dense. But it equals

$$\left(\bigcap_n U_n\right) \cap \left(\bigcap_n V_n\right) = \mathbb{Q} \cap (\mathbb{R} \setminus \mathbb{Q}) = \varnothing,$$

which is not dense in the nonempty space $\mathbb{R}$.

Take complements for the equivalent formulation.

If $\mathbb{R} \setminus \mathbb{Q} = \bigcup_n F_n$ with each F_n closed, De Morgan's law would give $\mathbb{Q} = \bigcap_n (\mathbb{R} \setminus F_n)$ with open terms.

Conclusion. Neither proposed representation is possible.

21.12 Exercise 21.12: Infinitely many disjoint sets of second category

What we must show. Give pairwise disjoint subsets of $\mathbb{R}$, each not a countable union of nowhere dense subsets of $\mathbb{R}$.

Choose intervals and invoke the right Baire consequence.

For $n \geq 1$, let $E_n = (2n, 2n+1)$. These intervals are pairwise disjoint. Suppose some E_n were a countable union of nowhere dense sets. By Theorem 21.7(c), its complement would be dense in $\mathbb{R}$. But that complement misses the nonempty open interval E_n, contradicting density.

Conclusion. The sequence $((2n, 2n+1))_{n\geq 1}$ works. The category here is category in the ambient space $\mathbb{R}$, as the exercise requires.

21.13 Exercise 21.13: Oscillation detects continuity

What we must show. For

$$\omega_f(x) = \inf_{\delta>0} \sup\{|f(y) - f(z)| : y, z \in (x-\delta, x+\delta)\},$$

prove f is continuous at x exactly when $\omega_f(x) = 0$. The suprema may be infinite when f is locally unbounded; all quantities are nonnegative.

Continuity makes the oscillation zero.

Fix $\varepsilon > 0$. Continuity gives $\delta > 0$ such that $|f(y) - f(x)| < \varepsilon/4$ for $|y - x| < \delta$. For two such points, the triangle inequality gives $|f(y) - f(z)| < \varepsilon/2$. Taking the supremum can change a strict inequality into a nonstrict one, but still gives a bound at most $\varepsilon/2 < \varepsilon$. Therefore $0 \leq \omega_f(x) < \varepsilon$. Since this holds for every $\varepsilon > 0$, $\omega_f(x) = 0$.

Zero oscillation gives continuity.

If the nonnegative infimum is zero, then for every $\varepsilon > 0$ some $\delta > 0$ has the corresponding supremum less than ε. Set $z = x$, which is permitted because x belongs to the interval. Then

$$|y - x| < \delta \implies |f(y) - f(x)| \leq \sup_{u,v\in(x-\delta,x+\delta)} |f(u) - f(v)| < \varepsilon.$$

Conclusion. This is continuity at x. It is important that the interval in the oscillation definition includes its center.

21.14 Exercise 21.14: Reduce every interval to the bounded closed case

What we must show. Complete Theorem 21.11 after its proof for $[a, b]$.

Assume a forbidden decomposition of an arbitrary interval.

Suppose a nondegenerate interval I is a disjoint union of at least two nondegenerate closed intervals J_α. Choose two distinct members. Select an interior point x of one and an interior point y of the other; relabel so $x < y$. Because I is an interval, $[x, y] \subseteq I$.

Intersect every piece with $[x, y]$.

Discard empty intersections. The remaining $K_\alpha = J_\alpha \cap [x, y]$ are pairwise disjoint closed intervals whose union is $[x, y]$. The interval containing x has positive length to the right of x, and the interval containing y has positive length to the left of y, because x, y were chosen in interiors. Every other surviving interval lies strictly between x and y and retains its original positive length. Thus none of the surviving intersections is a singleton. There are at least two of them, since the intervals containing x and y are distinct.

Use the case already established.

We have written the bounded closed interval $[x, y]$ as a disjoint union of at least two nondegenerate closed intervals. This contradicts the bounded closed case of Theorem 21.11.

Conclusion. The theorem holds for all nondegenerate intervals, including open, half-open, and unbounded intervals. The interior-point choice prevents unwanted singleton pieces at the truncation endpoints.